



MSc Thesis for Health Data Science

A Bayesian Time-Series Framework for Comparing Hospital Pressure Across Irish HSE Health Regions

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Declaration

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Master of Science is entirely our own work and had not been taken from the work of others or Artificial Intelligence, save and to the extent that such work has been cited and acknowledged within the text of our work.

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1 Introduction

2 Data

2.1 Data Collection

3 Methods

3.1 Data Collection Pipeline



Figure 1: Data pipeline from source to model input.

3.2 Model Development

Models were developed iteratively with increasing complexity as described in Gelman (2004), and visualised in figure 2. Model components were tested based on a priori expectation, and residual patterns. For instance: On trolley counts vary over time, so autoregressive components were first modeled. Then tested against each other on which minimised the Deviance Information Criterion (DIC) and residuals. Table 1 reads as a record of this model selection. Below is the components added to the models. The full model definitions can be found in Appendix 7.2.

Base Likelihood

All models share the same Normal likelihood:

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t}, \tau_i^{-1})$$

$$\tau_i \sim \Gamma(0.001, 0.001)$$

Where i and t define the region and week since January 1st 2023 respectively. The models differ in the mean function $\mu_{i,t}$, the prior structure on α_i , and autoregressive component, ϕ .

AR structure

Autoregressive component of lag 1 and 2 were initially added to the mean of the model. AR(1):

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + z_{i,t}, \tau_i^{-1}), \quad t > 1$$

where $z_{i,t} = \phi(y_{i,t-1} - \mu_{i,t-1})$

with $\phi \sim \text{Uniform}(-1, 1)$

AR(2):

$$\phi_1(y_{i,t-1} - \mu_{i,t-1}) + \phi_2(y_{i,t-2} - \mu_{i,t-2})$$

with $\phi_1, \phi_2 \sim \text{Uniform}(-1, 1)$

Cycles

Three cycles, 8 week, 26 week and 52 weeks, were tested alone, and together. A single cycle with period P (weeks) is defined as:

$$\text{cycle}_i(t; P) = \beta_i \cos\left(\frac{2\pi t}{P}\right) + \gamma_i \sin\left(\frac{2\pi t}{P}\right)$$

Where $P \in \{52, 26, 8\}$.

Dual cycle models sum the best performing 52 week cycle with the 8 and 26 week cycles:

$$\text{cycles}_{i,t} = \text{cycle}_{1,i}(t; P_1) + \text{cycle}_{2,i}(t; P_2)$$

Where $(P_1, P_2) \in \{(52, 26), (52, 8)\}$.

Cycle coefficients share the same weakly informative priors across all models:

$$\beta_i \sim \mathcal{N}(0, 0.001), \quad \gamma_i \sim \mathcal{N}(0, 0.001)$$

Event effects

The New Years effect was captured by fitting independent components to the 51st and 52nd weeks of each year, and the 1st week of each new year. The Mid West Region specific reset was modeled with three independent components on the 86th, 87th, and 88th weeks since January 1st, 2023 (the same as the 34th, 35th and 36th weeks of 2024).

Global New Year effect:

$$\begin{aligned} \text{ny}_t &= \delta_{\text{pre}} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid}} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post}} \cdot \mathbf{1}(t \bmod 52 = 2) \\ \delta_{\text{pre}}, \delta_{\text{mid}}, \delta_{\text{post}} &\sim \mathcal{N}(0, 0.001) \end{aligned}$$

Regional (i) New Year effect:

$$\begin{aligned} \text{ny}_{i,t} &= \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2) \\ \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \end{aligned}$$

Mid West reset effect:

$$\begin{aligned} \text{mwr}_{i=\text{mid west},t} &= \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \end{aligned}$$

Hierarchical intercepts

Replaces the independent priors on α_i with a hierarchical structure:

$$\alpha_i \sim \mathcal{N}(\mu_\alpha, \tau_\alpha^{-1})$$

with hyperpriors $\mu_\alpha \sim \mathcal{N}(0, 0.001)$ and $\tau_\alpha \sim \Gamma(0.001, 0.001)$.

3.3 Response Scaling

The on trolley counts were standardised to 10,000 people in each health region, Raw trolley counts vary across regions primarily because of population size rather than differences in hospital pressure. To enable fair comparison, counts were standardised to a rate per 10,000 catchment population.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region catchment population}_i} \times 10,000$$

This transforms the outcome into patients on trolley per 10,000 population per week, removing the confounding effect of region size and allowing direct comparison of hospital pressure across the six HSE health regions.

3.4 Posterior Sampling and Bayesian Ranking

Posterior sampling was performed using *Just another Gibbs sampler (JAGs)* (Plummer, 2003), implemented by pyJAGS (Nowotny, 2024), a Python interface to JAGS. Health region baselines, α_j^s , were used for ranking. Where j is one of the 6 HSE health regions, $j \in \{1, \dots, 6\}$, and s is the draw index, for $s \in \{1, \dots, n\}$ and total samples is defined as $n = 20,000$.

$$A = \begin{bmatrix} \alpha_1^1 & \cdots & \alpha_j^1 & \cdots & \alpha_6^1 \\ \vdots & & \vdots & & \vdots \\ \alpha_1^s & \cdots & \alpha_j^s & \cdots & \alpha_6^s \\ \vdots & & \vdots & & \vdots \\ \alpha_1^n & \cdots & \alpha_j^n & \cdots & \alpha_6^n \end{bmatrix}$$

Ranks follow the posterior ranking approach of Laird & Louis (1989). Where each sample is posterior sample is ranked to form rank distribution, R .

$$R = \begin{bmatrix} r_1^1 & \cdots & r_j^1 & \cdots & r_6^1 \\ \vdots & & \vdots & & \vdots \\ r_1^s & \cdots & r_j^s & \cdots & r_6^s \\ \vdots & & \vdots & & \vdots \\ r_1^n & \cdots & r_j^n & \cdots & r_6^n \end{bmatrix}$$

Where $r_j^s = \text{rank}(\alpha_j^s) = \sum_{i=1}^6 I(\alpha_j^s > \alpha_i^s)$. Given a tie, ranks are increased by $\frac{1}{2}$ steps for each tied value:

$$r_j^s = \sum_{i=1}^6 [\mathbf{1}(\alpha_j^s > \alpha_i^s) + \frac{1}{2}\mathbf{1}(\alpha_j^s = \alpha_i^s)]$$

Then posterior rank averages, $\hat{r}_i$, are calculated taking the mean across samples.

$$\hat{r}_j = \frac{1}{n} \sum_{s=1}^n r_j^s$$

$$\hat{R} = [\hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6]$$

Then posterior ranked averages are ranked once again to achieve ranked posterior ranked averages are calculated.

$$r_j^* = \text{rank}(\alpha_j^*) = \sum_{i=1}^6 I(\alpha_j^* > \alpha_i^*)$$

$$R^* = [r_1^*, r_2^*, r_3^*, r_4^*, r_5^*, r_6^*]$$

Where $r_j^* = \text{rank}(\hat{r}_j) = \sum_{i=1}^6 I(\hat{r}_j > \hat{r}_i)$.

4 Results

4.1 Model Comparison

Model fit was compared using the DIC. A DIC table was generated at every response scale.

4.1.1 Scaled by 2022 health region population (per 10,000 people)

4.1.2 Regional Baseline

4.1.3 Week-to-Week Autocorrelation

4.1.4 Annual Cycle

4.1.5 Regional Rankings

Table 1: DIC comparison across the models with a trolley rate per 10,000 people. Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v3.1	V2.2 + Mid West Reset	4555.9	93.6	4649.5
v4.1	V3.1 + Partial Pooling	4556.2	94.1	4650.3
v2.2	V1.1 + Regional New Year	4604.6	87.3	4691.9
v2.1	V1.1 + Global New Year	4682.7	57.5	4740.2
v2.3	V1.1 + Mid West Reset	5259.2	57.2	5316.4
v1.1	V0.1 + Annual Cycle	5294.9	51.2	5346.1
v0.2	V0.1 + AR(2)	5331.5	28.6	5360.1
v1.5	V1.1 + 8-Week Cycle	5284.9	75.3	5360.2
v0.1	AR(1) Only	5344.1	26.5	5370.6
v1.4	V1.1 + 6-Month Cycle	5301.3	75.8	5377.1
v1.3	V0.1 + 8-Week Cycle	5335.3	50.9	5386.1
v1.2	V0.1 + 26-Week Cycle	5349.4	51.1	5400.5

5 Discussion

6 References

- Gelman, A. (2004). Exploratory Data Analysis for Complex Models. *Journal of Computational and Graphical Statistics*, 13(4), 755–779. <https://doi.org/10.1198/106186004X11435>
- Laird, N. M., & Louis, T. A. (1989). Empirical Bayes ranking methods. *Journal of Educational Statistics*, 14(1), 29–46. <https://doi.org/10.3102/10769986014001029>
- Nowotny, M. (2024). *pyJAGS: A Python interface to JAGS* (Version 2.4.0) [Computer software]. <https://github.com/michaelnowotny/pyjags>
- Plummer, M. (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*. <https://www.r-project.org/conferences/DSC-2003/Proceedings/Plummer.pdf>

7 Appendix

7.1 Model Inheritance

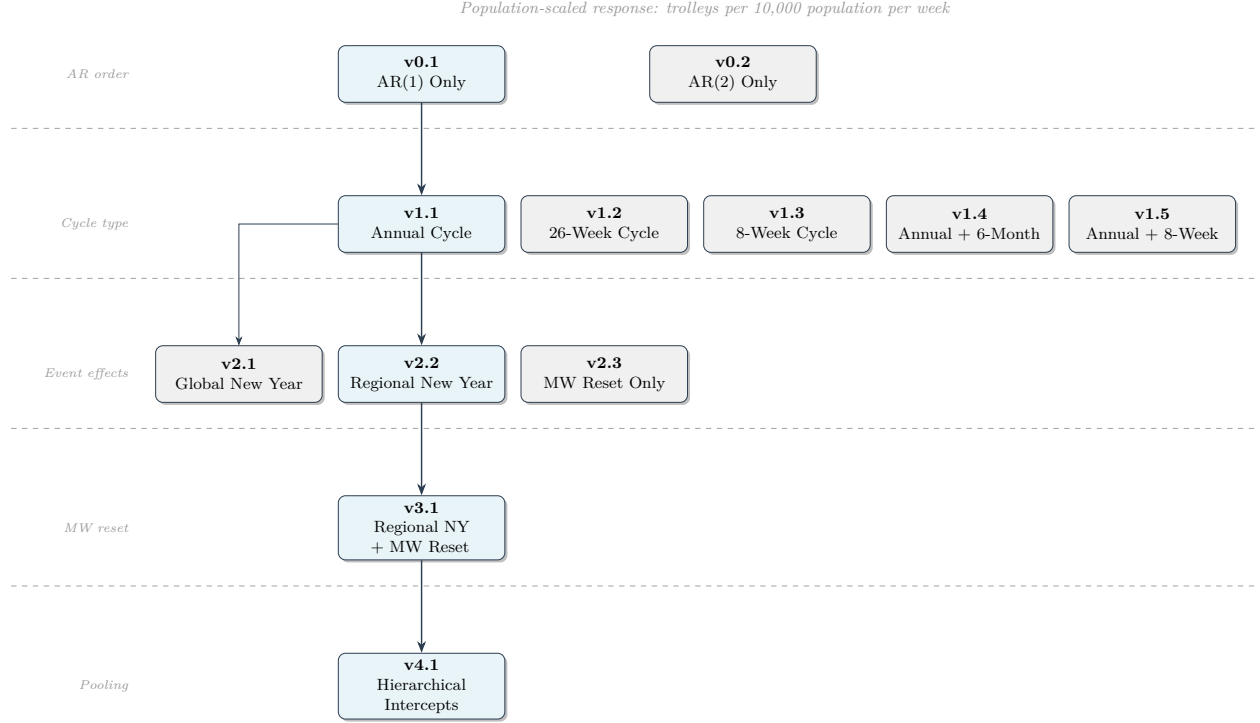


Figure 2: Model inheritance diagram. Spine nodes (blue) form the selected model sequence; grey nodes are tested variants. Thick arrows indicate the selected path; thin arrows indicate branches.

7.2 Full Model Specifications

Let $y_{i,t}$ denote the scaled trolley count for region i at week t , for $i \in \{1, \dots, 6\}$ and $t \in \{1, \dots, T\}$.

Model v0.1

AR(1) component with a baseline. Constant region-level intercept, no cycle, no event effects.

Likelihood

$$y_{i,1} \sim \mathcal{N}(\alpha_i, \tau_i^{-1})$$

$$y_{i,t} \sim \mathcal{N}(\alpha_i + \phi(y_{i,t-1} - \alpha_i), \tau_i^{-1}), \quad t > 1$$

Mean function

$$\mu_{i,t} = \alpha_i$$

Priors

$$\alpha_i \sim \mathcal{N}(0, 0.001)$$

$$\tau_i \sim \Gamma(0.001, 0.001)$$

$$\phi \sim \text{Uniform}(-1, 1)$$

Model v0.2

AR(2) baseline. Extends v0.1 with a second autoregressive lag.

Likelihood

$$\begin{aligned} y_{i,1} &\sim \mathcal{N}(\alpha_i, \tau_i^{-1}) \\ y_{i,2} &\sim \mathcal{N}(\alpha_i + \phi_1 (y_{i,1} - \alpha_i), \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\alpha_i + \phi_1 (y_{i,t-1} - \alpha_i) + \phi_2 (y_{i,t-2} - \alpha_i), \tau_i^{-1}), \quad t > 2 \end{aligned}$$

Mean function

$$\mu_{i,t} = \alpha_i$$

Priors

$$\begin{aligned} \alpha_i &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi_1, \phi_2 &\sim \text{Uniform}(-1, 1) \end{aligned}$$

Model v2.1

Extends v0.1. Adds a region-specific 52-week (annual) cycle.

Likelihood

$$\begin{aligned} y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi (y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1 \end{aligned}$$

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right)$$

Priors

$$\begin{aligned} \alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1) \end{aligned}$$

Model v2.2

Extends v0.1. Adds a region-specific 26-week (semi-annual) cycle.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{26}\right) + \gamma_i \sin\left(\frac{2\pi t}{26}\right)$$

Likelihood and priors identical to v2.1.

Model v2.3

Extends v0.1. Adds a region-specific 8-week cycle (pre-computed harmonic vectors).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos_8[t] + \gamma_i \sin_8[t]$$

where $\cos_8[t] = \cos(2\pi t/8)$ and $\sin_8[t] = \sin(2\pi t/8)$.

Likelihood and priors identical to v2.1.

Model v2.4

Extends v2.1. Adds a second harmonic at 26 weeks (dual cycle: annual + semi-annual).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_{1,i} \cos\left(\frac{2\pi t}{52}\right) + \gamma_{1,i} \sin\left(\frac{2\pi t}{52}\right) + \beta_{2,i} \cos_{26}[t] + \gamma_{2,i} \sin_{26}[t]$$

Priors

$$\alpha_i, \beta_{1,i}, \gamma_{1,i}, \beta_{2,i}, \gamma_{2,i} \sim \mathcal{N}(0, 0.001)$$

$$\tau_i \sim \Gamma(0.001, 0.001)$$

$$\phi \sim \text{Uniform}(-1, 1)$$

Model v2.5

Extends v2.1. Adds a second harmonic at 8 weeks (dual cycle: annual + 8-week).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_{1,i} \cos\left(\frac{2\pi t}{52}\right) + \gamma_{1,i} \sin\left(\frac{2\pi t}{52}\right) + \beta_{2,i} \cos_8[t] + \gamma_{2,i} \sin_8[t]$$

Priors identical to v2.4.

Model v3.1

Extends v2.1. Adds a global (shared across all regions) three-week New Year effect.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) + \delta_{\text{pre}} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid}} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post}} \cdot \mathbf{1}(t \bmod 52 = 2)$$

Priors

$$\alpha_i, \beta_i, \gamma_i \sim \mathcal{N}(0, 0.001)$$

$$\tau_i \sim \Gamma(0.001, 0.001)$$

$$\phi \sim \text{Uniform}(-1, 1)$$

$$\delta_{\text{pre}}, \delta_{\text{mid}}, \delta_{\text{post}} \sim \mathcal{N}(0, 0.001)$$

Model v3.2

Extends v3.1. New Year effect is region-specific rather than global.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) + \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2)$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i, \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v3.3

Extends v2.1. Adds a global Mid West funding reset effect (no New Year effect).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) + \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i$$

where $\text{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$.

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001)\end{aligned}$$

Model v4.1

Extends v3.2. Combines region-level New Year effects with the global Mid West funding reset.

Mean function

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2) \\ &\quad + \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i\end{aligned}$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i, \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001)\end{aligned}$$

7.2.1 Model v5.1

Let $y_{i,t}$ denote the scaled trolley count for region i at week t , for $i \in \{1, \dots, 6\}$ and $t \in \{1, \dots, T\}$.

Likelihood

$$y_{i,1} \sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1})$$

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1$$

Mean function

$$\begin{aligned} \mu_{i,t} = & \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ & + \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2) \\ & + \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i \end{aligned}$$

Model v5.1 mean function

Where $\text{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$. The β_i and γ_i coefficients capture the annual seasonal cycle, $\frac{\gamma_i}{\beta_i}$. The $\delta_{\text{pre}/\text{mid}/\text{post},i}$ components are region level new year effects. The σ component captures the Mid West funding reset.

The term $\phi(y_{i,t-1} - \mu_{i,t-1})$ in Equation (7.2.1) adds an AR(1) component, allowing each weeks observation to depend on the previous week's derivation from the mean.

Priors

The baseline parameters α_i are given a hierarchical prior, treating the six HSE regions as exchangeable within the same national health system:

$$\begin{aligned} \alpha_i & \sim \mathcal{N}(\mu_\alpha, \tau_\alpha^{-1}) \\ \mu_\alpha & \sim \mathcal{N}(0, 0.001) \\ \tau_\alpha & \sim \Gamma(0.001, 0.001) \end{aligned}$$

Remaining priors are independent and uninformative:

$$\begin{aligned} \beta_i, \gamma_i, \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} & \sim \mathcal{N}(0, 0.001) \\ \tau_i & \sim \Gamma(0.001, 0.001) \\ \phi & \sim \text{Uniform}(-1, 1) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} & \sim \mathcal{N}(0, 0.001) \end{aligned}$$

7.3 Alternative Response Scales

The primary analysis uses population-scaled trolley counts (per 10,000 catchment population). Two sensitivity scalings were also fitted to assess robustness of model selection and regional rankings to the choice of denominator.

7.3.1 Budget scaling

Weekly trolley counts scaled by the 2026 HSE regional budget allocation, expressing the outcome as trolleys per €1 billion allocated to the region.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region budget}_i / 10^6}$$

Table 2: DIC comparison across the models (trolley count scaled by 2026 regional budget). Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v4.1	V3.1 + Partial Pooling	19858.4	79.8	19938.1
v2.1	V1.1 + Global New Year	19920.3	54.9	19975.3
v3.1	V2.2 + Mid West Reset	19899.9	79.4	19979.3
v2.2	V1.1 + Regional New Year	19908.3	77.4	19985.7
v2.3	V1.1 + Mid West Reset	20169.5	50.4	20219.9
v1.1	V0.1 + Annual Cycle	20176.0	48.6	20224.6
v1.5	V1.1 + 8-Week Cycle	20171.4	72.1	20243.5
v1.4	V1.1 + 6-Month Cycle	20195.4	70.6	20266.0
v0.1	AR(1) Only	20253.9	27.0	20280.9
v0.2	V0.1 + AR(2)	20253.9	29.3	20283.2
v1.3	V0.1 + 8-Week Cycle	20248.9	50.6	20299.5
v1.2	V0.1 + 26-Week Cycle	20274.2	48.8	20323.0

Table 3: DIC comparison across models (response scaled by June 2025 health region inpatient bed count). Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v3.1	V2.2 + Mid West Reset	11448.9	93.0	11541.9
v4.1	V3.1 + Partial Pooling	11449.1	93.3	11542.4
v2.2	V1.1 + Regional New Year	11470.0	87.3	11557.3
v2.1	V1.1 + Global New Year	11510.1	57.2	11567.3
v2.3	V1.1 + Mid West Reset	11738.6	56.7	11795.3
v1.1	V0.1 + Annual Cycle	11756.7	51.0	11807.7
v1.5	V1.1 + 8-Week Cycle	11751.8	75.1	11826.9
v1.4	V1.1 + 6-Month Cycle	11773.3	75.3	11848.6
v0.1	AR(1) Only	11828.7	26.6	11855.3
v0.2	V0.1 + AR(2)	11828.3	28.6	11856.8
v1.3	V0.1 + 8-Week Cycle	11823.4	50.7	11874.1
v1.2	V0.1 + 26-Week Cycle	11845.5	50.9	11896.4

where health region budget_{*i*} is the 2026 budget allocation for region *i* in €thousands. This controls for funding differentials: a region receiving a larger allocation is expected to handle a proportionally larger patient volume.

7.3.2 Bed scaling

Weekly trolley counts scaled by the number of inpatient beds in each region, expressing the outcome as an overcapacity rate: for every 100 inpatient beds, how many patients were waiting on trolleys that week.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region inpatient beds}_i} \times 100$$

where health region inpatient beds_{*i*} is the total inpatient bed count across hospitals in region *i*, summed from the June 2025 bed census. Hospitals with missing bed counts are excluded from the regional sum.

7.4 Residual Diagnostics

[Residual diagnostic plots for each model.]

7.5 Posterior Mean Fit

[Posterior mean $\mu_{i,t}$ overlaid on observed $y_{i,t}$ for each spine model.]

7.6 Gelman-Rubin Convergence Summary

[Gelman-Rubin $\hat{R}$ summary table for each spine model.]

7.7 Posterior Parameter Estimates

[Posterior means and 95% credible intervals for each spine model.]